DSCI 552 Homework 3

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1. Time Series Classification Part 1: Feature Creation/Extraction

1(a) AReM dataset

The supplied ZIP is used directly: 88 recordings across seven activities, each containing six sensor time series. Original CSVs are preserved. Every file and numeric row was validated; 42,239 observed rows were retained.

1(b) Train/test split

For bending1 and bending2, dataset1.csv and dataset2.csv are test instances. For every other activity, dataset1.csv through dataset3.csv are test instances. All remaining recordings are training instances.

Activity	Training	Test	Total
bending1	5	2	7
bending2	4	2	6
cycling	12	3	15
lying	12	3	15
sitting	12	3	15
standing	12	3	15
walking	12	3	15
Total	69	19	88

Reasoning: 7 + 6 + 5(15) = 88 total; 2 + 2 + 5(3) = 19 test; (7 - 2) + (6 - 2) + 5(15 - 3) = 69 training. Whole recordings stay together. IDs follow alphabetical activity order, then numeric dataset order.

1(c)(i) Common time-domain features

Common features include minimum, maximum, range, mean, median, quantiles, variance, standard deviation, interquartile range, skewness, kurtosis, energy, root mean square, successive absolute changes, slope, level crossings, peaks, and lag autocorrelation. Distribution summaries measure level and spread; change and correlation features retain temporal-order information. [1, 2]

1(c)(ii) Extracted features and cleaning

Extract minimum, maximum, mean, median, sample standard deviation (ddof = 1), first quartile, and third quartile from each channel. Quartiles use linear interpolation. The resulting numeric dataframe is **88 × 42**, without normalization. All values appear in the appendix and in results/extracted_features.csv.

Channel order: 1 avg_rss12; 2 var_rss12; 3 avg_rss13; 4 var_rss13; 5 avg_rss23; 6 var_rss23. Columns are min1, max1, mean1, median1, std1, q1_1, q3_1, and similarly for channels 2-6.

Cleaning: bending2/dataset4.csv has 480 whitespace-separated rows. cycling/dataset9.csv and cycling/dataset14.csv each have one extra trailing comma. These delimiters are repaired during parsing; no observed row is dropped. sitting/dataset8.csv has 479 observations: timestamp 13,500 ms is absent in the supplied archive. Its summaries use the 479 observed values without imputation. The other 87 files each contain 480 observations.

[1] tsfresh, Feature overview: tsfresh.readthedocs.io/en/latest/text/list_of_features.html
[2] catch22 authors, mean and spread extensions: github.com/DynamicsAndNeuralSystems/catch22

1(c)(ii) Definitions and reasoning

A recording is the classification unit because it represents one performed activity. Treating timestamps as separate instances would change the task and inflate the sample size. Each of its six channels contributes seven summaries, producing 6 × 7 = 42 numeric columns.

For observed values x1, ..., xn in one channel:

$$\bar{x} = \frac{1}{n}\sum_{t=1}^{n} x_t, \qquad s = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}(x_t - \bar{x})^2}$$

Type	Meaning and reasoning
Minimum / maximum	Observed lower / upper bounds; summarize extremes.
Mean	Typical level using every observation.
Median	Central location less sensitive to extreme values.
Standard deviation	Within-recording spread around the mean.
First / third quartile	Lower / upper portions of the central distribution.

For linear-interpolated quantiles at probability p, sort the values and index them from zero. Set h = (n - 1)p, k = floor(h), and a = h - k. The quantile is (1 - a)x(k) + ax(k + 1). Use p = 0.25 and 0.75 for quartiles and p = 0.5 for the median.

The n - 1 denominator specifies the conventional sample standard deviation. Because neighboring time points may be dependent, this convention does not guarantee unbiased variance estimation for these time series.

Why raw values? Signal level and variability may help distinguish activities. Standardizing each recording separately would force its mean to zero and standard deviation to one, eliminating these differences. Across-recording feature standardization is unnecessary for the descriptive summaries and scale-independent activity-separation ratios used here.

Why these feature families? Level, spread, and extremes can reflect different aspects of activity. Temporal-order features such as successive changes and autocorrelation can add dynamics, but they are outside the seven requested summaries. A permutation of the observed time points leaves all seven extracted summaries unchanged, so they do not preserve the complete sequence shape.

Why preserve the shorter recording? Repairing delimiters changes formatting without changing measurements. Imputing the missing row would add an unobserved value and could alter extrema, quantiles, and spread. Using the 479 available values avoids inventing data; the difference from an unobserved complete recording is unknown.

1(c)(iii) Reasoning for the bootstrap

The sampling distribution of a standard deviation can be asymmetric and depends on the feature distribution. Resampling whole recordings approximates its sampling variability without imposing normality on the feature values. Keeping the central 90% leaves 5% in each tail; 10,000 draws reduce Monte Carlo variability, and seed 552 ensures reproducibility. Whole-row sampling also retains dependence among feature columns. The next page reports all endpoints.

Interpret intervals under an independent-recording approximation, conditional on the observed activity mixture. For example, min2 is zero in every observed recording, so its standard deviation and interval are all zero. This observed degeneracy does not guarantee zero variation in every future recording.

1(c)(iii) Feature standard deviations: 90% bootstrap intervals

For each feature, the point estimate is its sample standard deviation across all 88 recordings (ddof = 1). Use a nonparametric percentile bootstrap: draw 88 whole instance rows with replacement, compute all 42 standard deviations, and repeat 10,000 times with NumPy default_rng seed 552. The interval is the 5th to 95th percentile, using linear interpolation. All 42 requested features are reported below; values are rounded to four decimals.

feature	estimated_std	ci_lower_90	ci_upper_90
min1	9.5700	8.2622	10.7614
max1	4.3944	3.3280	5.2849
mean1	5.3357	4.7040	5.8864
median1	5.4401	4.7903	6.0043
std1	1.7722	1.5722	1.9500
q1_1	6.1536	5.5816	6.6524
q3_1	5.1389	4.3319	5.8409
min2	0.0000	0.0000	0.0000
max2	5.0627	4.6228	5.4022
mean2	1.5742	1.3941	1.7068
median2	1.4122	1.2361	1.5440
std2	0.8841	0.8026	0.9418
q1_2	0.9464	0.8294	1.0359
q3_2	2.1253	1.8914	2.2967
min3	2.9565	2.7601	3.1079
max3	4.8751	4.1702	5.4573
mean3	4.0084	3.4163	4.4918
median3	4.0364	3.4388	4.5351
std3	0.9467	0.7614	1.1198
q1_3	4.2207	3.6305	4.7049
q3_3	4.1716	3.5434	4.6940
min4	0.0000	0.0000	0.0000
max4	2.1836	1.9741	2.3595
mean4	1.1661	1.0740	1.2225
median4	1.1456	1.0522	1.2017
std4	0.4582	0.4207	0.4866
q1_4	0.8436	0.7726	0.8911
q3_4	1.5525	1.4300	1.6284
min5	6.1240	4.4033	7.5110
max5	5.7412	4.7406	6.5541
mean5	5.6756	4.4016	6.7385
median5	5.8138	4.4902	6.9103
std5	1.0249	0.8132	1.2158
q1_5	6.0965	4.7701	7.2176
q3_5	5.5317	4.3398	6.5283
min6	0.0458	0.0000	0.0785
max6	2.5189	2.2508	2.7639
mean6	1.1548	1.0601	1.2163
median6	1.0865	0.9925	1.1487
std6	0.5176	0.4781	0.5455
q1_6	0.7586	0.6882	0.8087
q3_6	1.5236	1.3985	1.6015

These are pointwise 90% intervals, not simultaneous coverage. They estimate between-recording dispersion under an independent-recording approximation, conditional on the observed activity mixture. They do not resample within-recording time points or establish new-subject generalization. A std1-std6 feature is a within-recording statistic; its estimated_std here is variation of that statistic across recordings. Large dispersion alone does not imply classification importance.

1(c)(iv) Three selected feature types

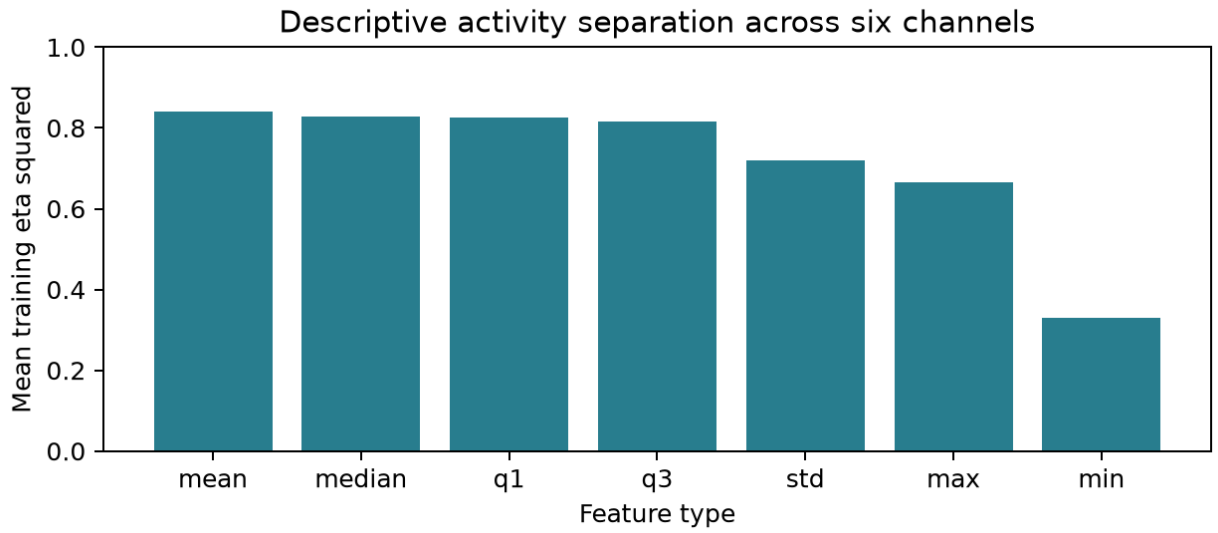
Choice: mean, standard deviation, and maximum, giving 18 channel-specific features. This judgment balances separation of activities and complementary information. It uses only the 69 training recordings and fits no classification models.

For each feature, compare the between-activity sum of squares with total variation:

$$\eta^2 = \frac{\sum_g n_g(\bar{z}_g - \bar{z})^2}{\sum_i (z_i - \bar{z})^2}$$

Average this scale-independent descriptive measure over the six channels. Higher values indicate more activity separation; they are not estimates of predictive accuracy.

Feature type	Mean training eta squared
mean	0.8406
median	0.8273
q1	0.8252
q3	0.8158
std	0.7199
max	0.6643
min	0.3287



Mean ranks first (0.8406), capturing typical signal level. **Standard deviation** captures fluctuation, with separation 0.7199 and average absolute correlation to mean of 0.6047. **Maximum** captures upper excursions, with separation 0.6643 and correlations of 0.7865 to mean and 0.6278 to standard deviation.

Median, q1, and q3 separate activities strongly, but their correlations with mean are 0.9907, 0.9796, and 0.9806, respectively. They largely repeat location information. Minimum has the weakest separation (0.3287), with some constant channels. Correlations are averaged over matching channels with positive variance in both columns; undefined constant-column correlations are excluded explicitly.

This is an interpretable, evidence-based judgment, not proof of an optimal triplet. The bootstrap table quantifies uncertainty in dispersion, whereas the training-only evidence directly addresses activity separation and redundancy.

2. ISLR 3.7.4

The supplied textbook gives n = 100 observations and one predictor. Compare ordinary least-squares linear and cubic models on the same training data. Assume independent test observations from the same population. The models are nested: the cubic model reduces to the linear model when its quadratic and cubic coefficients are zero.

$$\mathcal{L}: Y = \beta_0 + \beta_1 X + \varepsilon, \quad \mathcal{C}: Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$$

2(a) True relationship linear: training RSS

Cubic training RSS is no greater than linear training RSS. Minimizing over four coefficients includes the two-coefficient solution as a candidate:

$$\min_{\beta_0, \beta_1, \beta_2, \beta_3} \sum_i (y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2 - \beta_3 x_i^3)^2 \;\le\; \min_{\beta_0, \beta_1} \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$$

Under full rank, a true linear mean, and independent homoskedastic errors with variance $\sigma^2 > 0$, $E[RSS_L \mid X] = 98\sigma^2$ and $E[RSS_C \mid X] = 96\sigma^2$. Reasoning: $RSS = \varepsilon^T(I - H)\varepsilon$ and $E[RSS \mid X] = \sigma^2 \operatorname{trace}(I - H) = (n - p)\sigma^2$, where H has rank p. Extra parameters typically reduce training RSS by fitting noise; equality remains possible.

2(b) True relationship linear: test RSS

Expected test RSS is lower for linear regression under the usual assumptions. A particular finite test set can reverse the ordering. Both models contain the true mean, so extra cubic parameters add estimation variance without reducing bias:

$$E[(Y_0 - \hat{f}(x_0))^2 \mid x_0] = \sigma^2 + \text{Bias}[\hat{f}(x_0)]^2 + \text{Var}[\hat{f}(x_0)]$$

For a precise variance comparison, let X_L contain (1, x), Z contain (x^2, x^3), $A = (X_L^T X_L)^{-1} X_L^T Z$, and $W = Z - X_L A$. At a test point set $l_0 = (1, x_0)^T$, $z_0 = (x_0^2, x_0^3)^T$, and $w_0 = z_0 - A^T l_0$. Orthogonality of the residualized predictors gives:

$$\text{Var}(\hat{f}_C(x_0) \mid X) - \text{Var}(\hat{f}_L(x_0) \mid X) = \sigma^2 w_0^T (W^T W)^{-1} w_0 \ge 0$$

With full rank and error covariance $\sigma^2 I$, this extra term is nonnegative. Summing over equal-sized test sets gives the expected RSS ordering; equality can occur at points where the added variance is zero.

2(c) True relationship nonlinear: training RSS

Cubic training RSS is still no greater. The optimization inequality in 2(a) holds for any observed response vector, independent of the true regression function. The true function need not itself be cubic.

2(d) True relationship nonlinear: test RSS

There is not enough information to order expected test RSS. For m test observations:

$$\frac{E[RSS_C - RSS_L]}{m} = E_{X_0}[\text{Bias}_C^2 - \text{Bias}_L^2 + \text{Var}_C - \text{Var}_L]$$

Irreducible noise cancels. Substantial quadratic or cubic structure can reduce bias enough to offset added variance; a small departure from linearity may not. Nonlinearity can also lie outside both polynomial families. The true function, noise level, and predictor distribution determine the tradeoff, so either model can win.

Problem source: supplied ISLP textbook, printed page 128 (PDF page 137), Exercise 4.

1(c)(ii) Appendix: time series 1, instances 1-44

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min1	max1	mean1	median1	std1	q1_1	q3_1
1	bending1	dataset1.csv	37.2500	45.0000	40.6248	40.5000	1.4770	39.2500	42.0000
2	bending1	dataset2.csv	38.0000	45.6700	42.8128	42.5000	1.4355	42.0000	43.6700
3	bending1	dataset3.csv	35.0000	47.4000	43.9545	44.3300	1.5588	43.0000	45.0000
4	bending1	dataset4.csv	33.0000	47.7500	42.1798	43.5000	3.6707	39.1500	45.0000
5	bending1	dataset5.csv	33.0000	45.7500	41.6781	41.7500	2.2435	41.3300	42.7500
6	bending1	dataset6.csv	37.0000	48.0000	43.4550	43.2500	1.3861	42.5000	45.0000
7	bending1	dataset7.csv	36.2500	48.0000	43.9691	44.5000	1.6184	43.3100	44.6700
8	bending2	dataset1.csv	12.7500	51.0000	24.5630	24.2500	3.7375	23.1875	26.5000
9	bending2	dataset2.csv	0.0000	42.7500	27.4646	28.0000	3.5836	25.5000	30.0000
10	bending2	dataset3.csv	21.0000	50.0000	32.5862	33.0000	6.2381	26.1875	34.5000
11	bending2	dataset4.csv	27.5000	33.0000	29.8819	30.0000	1.1538	29.0000	30.2700
12	bending2	dataset5.csv	19.0000	45.5000	30.9381	29.0000	7.6841	26.7500	38.0000
13	bending2	dataset6.csv	25.0000	47.5000	31.0582	29.7100	4.8298	27.5000	31.8125
14	cycling	dataset1.csv	24.2500	45.0000	37.1770	36.2500	3.5813	34.5000	40.2500
15	cycling	dataset2.csv	28.7500	44.7500	37.5612	36.8750	3.2265	35.2500	40.2500
16	cycling	dataset3.csv	22.0000	44.6700	37.0587	36.0000	3.7102	34.5000	40.0625
17	cycling	dataset4.csv	19.0000	44.0000	36.2284	36.0000	3.5286	34.0000	39.0000
18	cycling	dataset5.csv	26.5000	44.3300	36.6873	36.0000	3.5294	34.2500	39.3725
19	cycling	dataset6.csv	25.3300	45.0000	37.1143	36.2500	3.7104	34.5000	40.2500
20	cycling	dataset7.csv	26.7500	44.7500	36.8634	36.3300	3.5558	34.5000	39.7500
21	cycling	dataset8.csv	26.2500	44.2500	36.9575	36.2900	3.4349	34.5000	40.2500
22	cycling	dataset9.csv	27.7500	44.6700	37.1448	36.3300	3.7589	34.0000	40.5000
23	cycling	dataset10.csv	27.0000	45.0000	36.8195	36.0000	3.9005	33.7500	40.2500
24	cycling	dataset11.csv	27.0000	44.3300	36.5417	36.0000	4.0189	33.2500	39.8125
25	cycling	dataset12.csv	18.5000	44.2500	35.7524	36.0000	4.6148	33.0000	39.3300
26	cycling	dataset13.csv	19.0000	43.7500	35.8799	36.0000	4.6149	33.0000	39.5000
27	cycling	dataset14.csv	23.3300	43.5000	36.2441	36.7500	3.8220	33.4575	39.2500
28	cycling	dataset15.csv	24.2500	45.0000	37.1770	36.2500	3.5813	34.5000	40.2500
29	lying	dataset1.csv	23.5000	30.0000	27.7164	27.5000	1.4423	27.0000	29.0000
30	lying	dataset2.csv	24.7500	48.3300	44.1829	48.0000	7.4956	48.0000	48.0000
31	lying	dataset3.csv	48.0000	48.2500	48.0042	48.0000	0.0320	48.0000	48.0000
32	lying	dataset4.csv	34.0000	51.0000	42.7061	40.5000	3.5375	40.2500	48.0000
33	lying	dataset5.csv	39.0000	41.0000	39.6678	39.5000	0.2802	39.5000	39.7500
34	lying	dataset6.csv	0.0000	40.6700	39.5062	39.5000	1.8175	39.5000	39.6700
35	lying	dataset7.csv	39.0000	40.0000	39.4338	39.5000	0.2086	39.3300	39.5000
36	lying	dataset8.csv	39.0000	40.0000	39.3471	39.3300	0.2314	39.2500	39.5000
37	lying	dataset9.csv	39.0000	56.2500	47.3251	42.6700	5.9613	42.0000	54.0000
38	lying	dataset10.csv	23.5000	30.0000	27.7164	27.5000	1.4423	27.0000	29.0000
39	lying	dataset11.csv	48.0000	48.2500	48.0042	48.0000	0.0320	48.0000	48.0000
40	lying	dataset12.csv	39.0000	41.0000	39.6678	39.5000	0.2802	39.5000	39.7500
41	lying	dataset13.csv	39.0000	40.0000	39.4338	39.5000	0.2086	39.3300	39.5000
42	lying	dataset14.csv	39.0000	56.2500	47.3251	42.6700	5.9613	42.0000	54.0000
43	lying	dataset15.csv	23.5000	30.0000	27.7164	27.5000	1.4423	27.0000	29.0000
44	sitting	dataset1.csv	33.2500	48.0000	42.3636	43.0000	2.0682	42.0000	43.5000

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 1, instances 45-88

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min1	max1	mean1	median1	std1	q1_1	q3_1
45	sitting	dataset2.csv	37.0000	50.7500	45.9177	45.6350	1.0890	45.0000	46.5000
46	sitting	dataset3.csv	44.5000	46.7500	45.2397	45.0000	0.4299	45.0000	45.3300
47	sitting	dataset4.csv	42.5000	46.0000	44.7540	45.0000	0.5203	44.7500	45.0000
48	sitting	dataset5.csv	25.5000	45.7500	36.5601	36.5000	5.9151	30.0000	42.5000
49	sitting	dataset6.csv	27.0000	38.6700	32.9060	30.2500	3.5058	30.0000	37.0000
50	sitting	dataset7.csv	27.5000	50.7500	38.8884	36.7500	3.6429	36.2500	42.0000
51	sitting	dataset8.csv	42.0000	45.0000	42.8220	42.5000	0.8918	42.0000	43.5000
52	sitting	dataset9.csv	41.7500	46.5000	43.1909	42.5000	1.2303	42.0000	44.5000
53	sitting	dataset10.csv	35.2500	48.5000	40.2193	39.2500	2.7412	37.7500	42.5000
54	sitting	dataset11.csv	28.5000	48.2500	43.8848	45.2500	3.1989	42.0000	46.5000
55	sitting	dataset12.csv	39.5000	45.0000	42.1116	42.0000	1.1222	41.5000	42.0000
56	sitting	dataset13.csv	39.6700	44.7500	42.2827	41.5000	1.3561	41.5000	44.3300
57	sitting	dataset14.csv	40.0000	44.6700	42.3602	42.0000	1.0174	41.5000	43.2500
58	sitting	dataset15.csv	29.2500	46.0000	42.7289	43.2500	2.0464	41.3300	44.5000
59	standing	dataset1.csv	33.3300	48.0000	44.3347	45.0000	2.4769	42.2500	46.5000
60	standing	dataset2.csv	35.5000	46.2500	43.1749	43.6700	1.9891	42.5000	44.5000
61	standing	dataset3.csv	32.7500	47.0000	42.7606	44.5000	3.3989	41.3300	45.3725
62	standing	dataset4.csv	30.0000	46.6700	42.6485	42.7500	2.3953	41.5000	45.0000
63	standing	dataset5.csv	36.0000	47.5000	43.7200	45.0000	2.3841	43.0000	45.0000
64	standing	dataset6.csv	34.5000	47.7500	44.4711	45.0000	1.7726	45.0000	45.2500
65	standing	dataset7.csv	35.5000	48.0000	46.2249	46.0000	1.7483	45.2500	48.0000
66	standing	dataset8.csv	29.7500	48.0000	46.9322	47.5000	1.8327	47.2375	47.7500
67	standing	dataset9.csv	36.3300	47.6700	45.3996	45.5000	1.3281	45.0000	46.3300
68	standing	dataset10.csv	36.0000	45.8000	42.4199	42.6700	2.5201	41.3300	44.6175
69	standing	dataset11.csv	37.0000	48.2500	42.5170	42.5000	2.1958	41.0000	44.5000
70	standing	dataset12.csv	36.2500	45.5000	42.9594	42.6700	1.5009	42.0000	44.3300
71	standing	dataset13.csv	36.0000	47.3300	42.6746	43.6700	2.3842	40.0000	44.7500
72	standing	dataset14.csv	36.2500	45.7500	43.1875	44.7500	2.4912	39.7500	45.0000
73	standing	dataset15.csv	36.0000	47.3300	44.4412	45.0000	2.4178	44.6275	45.7500
74	walking	dataset1.csv	19.3300	43.5000	34.2278	35.5000	4.8896	30.5000	37.7500
75	walking	dataset2.csv	12.5000	45.0000	33.5097	34.1250	4.8509	30.5000	36.7500
76	walking	dataset3.csv	15.0000	46.7500	34.6606	35.0000	5.3151	31.0000	38.2500
77	walking	dataset4.csv	18.0000	46.0000	35.1933	36.0000	4.7519	32.0000	38.7500
78	walking	dataset5.csv	20.7500	46.2500	34.7633	35.2900	4.7422	31.6700	38.2500
79	walking	dataset6.csv	21.5000	51.0000	34.9358	35.5000	4.6459	32.0000	38.0625
80	walking	dataset7.csv	18.3300	47.6700	34.3330	34.7500	4.9488	31.2500	38.0000
81	walking	dataset8.csv	18.3300	45.7500	34.5999	35.1250	4.7318	31.5000	38.0000
82	walking	dataset9.csv	15.5000	43.6700	34.2259	34.7500	4.4418	31.2500	37.2500
83	walking	dataset10.csv	21.5000	51.2500	34.2535	35.0000	4.9407	30.9375	37.7500
84	walking	dataset11.csv	19.5000	45.3300	33.5869	34.2500	4.6509	30.2500	37.0000
85	walking	dataset12.csv	19.7500	45.5000	34.3227	35.2500	4.7525	31.0000	38.0000
86	walking	dataset13.csv	19.5000	46.0000	34.5462	35.2500	4.8423	31.2500	37.8125
87	walking	dataset14.csv	23.5000	46.2500	34.8732	35.2500	4.5317	31.7500	38.2500
88	walking	dataset15.csv	19.2500	44.0000	34.4732	35.0000	4.7967	31.2500	38.0000

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 2, instances 1-44

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min2	max2	mean2	median2	std2	q1_2	q3_2
1	bending1	dataset1.csv	0.0000	1.3000	0.3586	0.4300	0.3226	0.0000	0.5000
2	bending1	dataset2.csv	0.0000	1.2200	0.3724	0.4700	0.2892	0.0000	0.5000
3	bending1	dataset3.csv	0.0000	1.7000	0.4263	0.4700	0.3387	0.0000	0.5000
4	bending1	dataset4.csv	0.0000	3.0000	0.6960	0.5000	0.6309	0.0000	1.1200
5	bending1	dataset5.csv	0.0000	2.8300	0.5360	0.5000	0.4055	0.4300	0.7100
6	bending1	dataset6.csv	0.0000	1.5800	0.3781	0.4700	0.3156	0.0000	0.5000
7	bending1	dataset7.csv	0.0000	1.5000	0.4131	0.4700	0.2631	0.4300	0.5000
8	bending2	dataset1.csv	0.0000	6.8700	0.5908	0.4300	0.8374	0.0000	0.7100
9	bending2	dataset2.csv	0.0000	7.7600	0.4497	0.4300	0.7672	0.0000	0.5000
10	bending2	dataset3.csv	0.0000	9.9000	0.5161	0.4300	0.7900	0.0000	0.8225
11	bending2	dataset4.csv	0.0000	1.0000	0.2564	0.0000	0.2758	0.0000	0.4700
12	bending2	dataset5.csv	0.0000	6.4000	0.4672	0.4300	0.7344	0.0000	0.5000
13	bending2	dataset6.csv	0.0000	6.3800	0.4055	0.4300	0.5667	0.0000	0.5000
14	cycling	dataset1.csv	0.0000	8.5800	2.3742	1.9200	1.6018	1.3000	3.1300
15	cycling	dataset2.csv	0.0000	9.9100	2.0807	1.7000	1.6393	1.1200	2.8700
16	cycling	dataset3.csv	0.0000	14.1700	2.4381	1.9200	1.9969	1.2500	3.3500
17	cycling	dataset4.csv	0.0000	12.2800	2.8317	2.2850	1.8867	1.5000	3.8450
18	cycling	dataset5.csv	0.0000	12.8900	2.9730	2.3600	2.0739	1.4100	4.2625
19	cycling	dataset6.csv	0.0000	10.8400	2.7300	2.2400	1.9244	1.3000	3.7175
20	cycling	dataset7.csv	0.0000	11.6800	2.7573	2.2300	2.1160	1.2200	4.0225
21	cycling	dataset8.csv	0.0000	8.6400	2.4201	2.0500	1.7249	1.2500	3.3500
22	cycling	dataset9.csv	0.0000	10.7600	2.4191	1.8800	1.6432	1.2500	3.3675
23	cycling	dataset10.csv	0.0000	10.4700	2.6001	2.1200	1.8214	1.3000	3.8700
24	cycling	dataset11.csv	0.0000	10.4300	2.8480	2.4500	1.8924	1.3000	4.1500
25	cycling	dataset12.csv	0.0000	12.6000	3.3281	2.8300	2.1406	1.7750	4.3300
26	cycling	dataset13.csv	0.0000	11.2000	3.4143	2.9200	2.2380	1.6550	4.8450
27	cycling	dataset14.csv	0.0000	9.7100	2.7360	2.1700	2.0920	1.2200	3.7400
28	cycling	dataset15.csv	0.0000	8.5800	2.3742	1.9200	1.6018	1.3000	3.1300
29	lying	dataset1.csv	0.0000	1.7900	0.3637	0.4300	0.3948	0.0000	0.5000
30	lying	dataset2.csv	0.0000	3.1100	0.1019	0.0000	0.3468	0.0000	0.0000
31	lying	dataset3.csv	0.0000	0.4300	0.0072	0.0000	0.0551	0.0000	0.0000
32	lying	dataset4.csv	0.0000	4.8500	0.5198	0.5000	0.4841	0.0000	0.8300
33	lying	dataset5.csv	0.0000	1.0000	0.5836	0.5000	0.1946	0.5000	0.7100
34	lying	dataset6.csv	0.0000	1.0000	0.4965	0.5000	0.1455	0.4700	0.5000
35	lying	dataset7.csv	0.0000	1.0000	0.4221	0.4700	0.1604	0.4300	0.5000
36	lying	dataset8.csv	0.0000	0.5000	0.3664	0.4700	0.1958	0.4300	0.5000
37	lying	dataset9.csv	0.0000	8.4900	0.2743	0.0000	0.5627	0.0000	0.4700
38	lying	dataset10.csv	0.0000	1.7900	0.3637	0.4300	0.3948	0.0000	0.5000
39	lying	dataset11.csv	0.0000	0.4300	0.0072	0.0000	0.0551	0.0000	0.0000
40	lying	dataset12.csv	0.0000	1.0000	0.5836	0.5000	0.1946	0.5000	0.7100
41	lying	dataset13.csv	0.0000	1.0000	0.4221	0.4700	0.1604	0.4300	0.5000
42	lying	dataset14.csv	0.0000	8.4900	0.2743	0.0000	0.5627	0.0000	0.4700
43	lying	dataset15.csv	0.0000	1.7900	0.3637	0.4300	0.3948	0.0000	0.5000
44	sitting	dataset1.csv	0.0000	4.4400	0.4926	0.5000	0.4036	0.4300	0.5000

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 2, instances 45-88

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min2	max2	mean2	median2	std2	q1_2	q3_2
45	sitting	dataset2.csv	0.0000	4.8700	0.3245	0.4300	0.3658	0.0000	0.5000
46	sitting	dataset3.csv	0.0000	1.0000	0.1730	0.0000	0.2512	0.0000	0.4300
47	sitting	dataset4.csv	0.0000	1.1200	0.1539	0.0000	0.2588	0.0000	0.4300
48	sitting	dataset5.csv	0.0000	12.1900	0.4222	0.0000	0.7730	0.0000	0.8200
49	sitting	dataset6.csv	0.0000	2.4900	0.4567	0.0000	0.5388	0.0000	1.0000
50	sitting	dataset7.csv	0.0000	6.0600	0.3834	0.4300	0.4889	0.0000	0.5000
51	sitting	dataset8.csv	0.0000	1.1200	0.3357	0.4300	0.2604	0.0000	0.5000
52	sitting	dataset9.csv	0.0000	1.5000	0.2663	0.4300	0.2716	0.0000	0.5000
53	sitting	dataset10.csv	0.0000	3.2800	0.6175	0.5000	0.4854	0.4300	1.0000
54	sitting	dataset11.csv	0.0000	3.2800	0.5174	0.5000	0.3941	0.4300	0.7100
55	sitting	dataset12.csv	0.0000	1.0900	0.3419	0.4700	0.2958	0.0000	0.5000
56	sitting	dataset13.csv	0.0000	1.0000	0.4984	0.5000	0.1480	0.4700	0.5000
57	sitting	dataset14.csv	0.0000	1.0000	0.4825	0.5000	0.2464	0.4300	0.5000
58	sitting	dataset15.csv	0.0000	4.7200	0.5553	0.5000	0.3825	0.4700	0.5000
59	standing	dataset1.csv	0.0000	3.9000	0.4330	0.4700	0.3706	0.0000	0.5000
60	standing	dataset2.csv	0.0000	2.1200	0.5066	0.5000	0.3074	0.4300	0.5000
61	standing	dataset3.csv	0.0000	3.3400	0.4862	0.4700	0.4465	0.0000	0.5000
62	standing	dataset4.csv	0.0000	2.9500	0.4028	0.4300	0.4137	0.0000	0.5000
63	standing	dataset5.csv	0.0000	1.9200	0.3667	0.4300	0.3741	0.0000	0.5000
64	standing	dataset6.csv	0.0000	2.1800	0.2905	0.0000	0.3929	0.0000	0.5000
65	standing	dataset7.csv	0.0000	4.5000	0.3124	0.0000	0.4645	0.0000	0.5000
66	standing	dataset8.csv	0.0000	4.6000	0.4297	0.5000	0.3673	0.4300	0.5000
67	standing	dataset9.csv	0.0000	1.6600	0.4601	0.5000	0.2699	0.4300	0.5000
68	standing	dataset10.csv	0.0000	2.1200	0.4606	0.4700	0.3874	0.0000	0.5050
69	standing	dataset11.csv	0.0000	2.1200	0.4407	0.4700	0.3494	0.0000	0.5000
70	standing	dataset12.csv	0.0000	2.6000	0.3529	0.4700	0.3136	0.0000	0.5000
71	standing	dataset13.csv	0.0000	2.1700	0.4192	0.4700	0.3684	0.0000	0.5000
72	standing	dataset14.csv	0.0000	2.8300	0.2713	0.0000	0.3288	0.0000	0.5000
73	standing	dataset15.csv	0.0000	4.5000	0.3466	0.4300	0.3581	0.0000	0.5000
74	walking	dataset1.csv	0.0000	14.5000	3.9957	3.6300	2.2711	2.3750	5.3425
75	walking	dataset2.csv	0.0000	13.0500	4.4508	4.0850	2.3387	2.7400	5.8900
76	walking	dataset3.csv	0.0000	13.4400	4.2009	3.9000	2.4802	2.3425	5.6725
77	walking	dataset4.csv	0.0000	16.2000	4.3210	3.8800	2.6255	2.4800	5.8700
78	walking	dataset5.csv	0.0000	12.6800	4.2238	3.9000	2.2780	2.4900	5.7225
79	walking	dataset6.csv	0.0000	12.2100	4.1158	3.8450	2.3352	2.3500	5.6250
80	walking	dataset7.csv	0.0000	12.4800	4.3970	3.9000	2.5687	2.4900	5.7450
81	walking	dataset8.csv	0.0000	15.3700	4.3988	4.0250	2.5385	2.5500	5.7675
82	walking	dataset9.csv	0.0000	17.2400	4.3545	3.9000	2.5190	2.4900	5.7650
83	walking	dataset10.csv	0.0000	13.5500	4.4579	4.1500	2.3872	2.7275	5.8900
84	walking	dataset11.csv	0.0000	14.6700	4.5766	4.2600	2.4611	2.8700	6.1050
85	walking	dataset12.csv	0.0000	13.4700	4.4563	3.9000	2.5993	2.4900	5.9700
86	walking	dataset13.csv	0.0000	12.4700	4.3720	4.1350	2.3604	2.4900	5.8000
87	walking	dataset14.csv	0.0000	14.8200	4.3806	3.9250	2.4354	2.5000	5.9325
88	walking	dataset15.csv	0.0000	13.8600	4.3593	3.9600	2.3887	2.4900	5.9000

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 3, instances 1-44

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min3	max3	mean3	median3	std3	q1_3	q3_3
1	bending1	dataset1.csv	4.0000	29.5000	19.0409	19.2500	4.4630	16.0000	23.2500
2	bending1	dataset2.csv	2.0000	29.5000	20.0970	21.0000	3.8937	19.0000	22.2500
3	bending1	dataset3.csv	6.5000	29.7500	22.1224	23.0000	3.0309	19.7500	24.0000
4	bending1	dataset4.csv	8.5000	30.0000	22.1836	23.0000	3.8105	20.5000	24.3725
5	bending1	dataset5.csv	3.0000	28.2500	19.0066	19.1250	4.0871	16.5000	22.0625
6	bending1	dataset6.csv	5.7500	27.0000	15.7933	15.0000	3.8476	13.0000	18.2700
7	bending1	dataset7.csv	1.5000	26.3300	15.8680	16.2500	3.7424	14.2500	18.0000
8	bending2	dataset1.csv	0.0000	25.3300	19.1213	20.2500	4.2996	18.0000	22.0000
9	bending2	dataset2.csv	7.5000	35.0000	20.8425	20.7500	3.8263	18.0000	22.5000
10	bending2	dataset3.csv	0.0000	28.2500	13.9416	14.2500	5.8833	10.0000	18.0000
11	bending2	dataset4.csv	0.0000	14.5000	8.1662	8.7500	3.4036	5.5000	11.2500
12	bending2	dataset5.csv	0.0000	32.7500	14.5898	15.7500	7.6389	9.6875	20.6900
13	bending2	dataset6.csv	2.0000	28.3300	15.2967	15.0000	4.8420	13.0000	17.7500
14	cycling	dataset1.csv	5.5000	26.7500	16.5311	16.6700	3.4309	14.2500	19.0000
15	cycling	dataset2.csv	6.5000	24.6700	16.5670	17.0000	3.6914	14.0000	19.5000
16	cycling	dataset3.csv	6.3300	24.0000	16.3883	16.5000	3.5380	13.7500	19.0000
17	cycling	dataset4.csv	5.5000	25.2500	15.4245	15.2500	3.2498	13.3300	17.5000
18	cycling	dataset5.csv	8.3300	28.2500	18.3968	18.0000	3.7711	15.6700	20.7500
19	cycling	dataset6.csv	3.0000	27.2500	16.6639	16.6700	3.6122	14.2500	18.8125
20	cycling	dataset7.csv	7.0000	27.0000	16.4947	16.0000	3.5882	14.0000	19.0000
21	cycling	dataset8.csv	7.0000	26.5000	15.3134	15.2500	3.0340	13.2500	17.3300
22	cycling	dataset9.csv	0.0000	24.7500	14.9985	15.0000	3.2895	13.0000	17.2500
23	cycling	dataset10.csv	5.6700	25.0000	15.3026	15.0000	3.5049	12.7500	17.6700
24	cycling	dataset11.csv	6.0000	27.6700	16.1561	16.0000	3.4390	14.0000	18.0500
25	cycling	dataset12.csv	5.0000	27.0000	16.0550	16.0000	3.4461	13.7300	18.3300
26	cycling	dataset13.csv	7.6700	26.5000	16.6904	17.0000	3.0094	14.7500	18.6900
27	cycling	dataset14.csv	7.0000	28.5000	18.4357	18.3300	4.0216	15.6700	21.0000
28	cycling	dataset15.csv	5.5000	26.7500	16.5311	16.6700	3.4309	14.2500	19.0000
29	lying	dataset1.csv	0.0000	13.2500	6.0786	6.2500	3.4690	3.5000	8.3300
30	lying	dataset2.csv	1.0000	16.5000	6.6800	6.2500	1.9365	5.6700	7.5000
31	lying	dataset3.csv	0.0000	13.0000	4.9006	5.5000	2.5664	3.0000	6.2500
32	lying	dataset4.csv	0.0000	23.0000	10.9100	12.0000	3.2084	9.0000	12.0000
33	lying	dataset5.csv	0.0000	15.0000	10.8342	12.0000	2.4695	10.2500	12.0000
34	lying	dataset6.csv	0.0000	13.2500	6.5501	7.0000	2.6071	4.5000	8.7500
35	lying	dataset7.csv	0.0000	12.2000	5.8411	6.0000	3.4979	3.0000	8.7500
36	lying	dataset8.csv	0.0000	11.3300	4.3822	4.5000	2.4568	2.5000	6.0000
37	lying	dataset9.csv	0.0000	18.2500	9.5319	10.7500	3.9871	6.0000	12.7500
38	lying	dataset10.csv	0.0000	13.2500	6.0781	6.2500	3.4687	3.5000	8.3300
39	lying	dataset11.csv	0.0000	13.0000	4.9006	5.5000	2.5664	3.0000	6.2500
40	lying	dataset12.csv	0.0000	15.0000	10.8342	12.0000	2.4695	10.2500	12.0000
41	lying	dataset13.csv	0.0000	12.2000	5.8411	6.0000	3.4979	3.0000	8.7500
42	lying	dataset14.csv	0.0000	18.2500	9.5319	10.7500	3.9871	6.0000	12.7500
43	lying	dataset15.csv	0.0000	13.2500	6.0786	6.2500	3.4690	3.5000	8.3300
44	sitting	dataset1.csv	3.0000	22.5000	17.1193	18.0000	3.1410	15.0000	19.3300

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 3, instances 45-88

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min3	max3	mean3	median3	std3	q1_3	q3_3
45	sitting	dataset2.csv	0.0000	24.0000	16.8715	18.7500	5.3599	15.0000	21.0000
46	sitting	dataset3.csv	2.0000	23.7500	12.8133	13.0000	4.1919	10.0000	15.2500
47	sitting	dataset4.csv	3.6700	27.2500	18.5796	19.2500	4.1281	17.0000	21.0000
48	sitting	dataset5.csv	1.0000	24.0000	14.8948	15.3300	5.4598	11.0000	20.2700
49	sitting	dataset6.csv	2.5000	23.7500	16.3059	18.5000	5.1649	12.0000	21.0000
50	sitting	dataset7.csv	1.5000	23.2500	14.8542	15.0000	3.9269	12.0000	18.0000
51	sitting	dataset8.csv	0.0000	23.0000	17.1997	18.0000	3.8980	16.0000	20.0000
52	sitting	dataset9.csv	0.0000	21.6700	14.8583	15.0000	4.7278	12.2500	18.4250
53	sitting	dataset10.csv	0.0000	23.5000	14.8512	16.0000	4.9292	11.9375	18.2500
54	sitting	dataset11.csv	0.0000	24.2500	14.9974	15.0000	5.6218	11.1875	18.7500
55	sitting	dataset12.csv	0.0000	21.8000	11.4422	12.0000	4.5397	9.0000	14.7500
56	sitting	dataset13.csv	0.0000	22.7500	9.4695	10.0000	5.3138	5.0000	14.0000
57	sitting	dataset14.csv	2.0000	23.5000	12.8845	12.7500	4.0129	11.4575	15.0000
58	sitting	dataset15.csv	0.0000	24.0000	15.8236	16.5000	4.8797	12.4575	20.0000
59	standing	dataset1.csv	0.0000	18.7500	11.6461	12.2500	3.6985	9.7500	14.5000
60	standing	dataset2.csv	1.0000	20.6700	12.7713	13.0000	3.2895	11.2500	15.0000
61	standing	dataset3.csv	1.5000	21.0000	15.0368	15.2500	3.1218	14.0000	17.0000
62	standing	dataset4.csv	3.0000	21.2500	18.1293	18.5000	2.8504	18.0000	19.7500
63	standing	dataset5.csv	6.6700	21.0000	17.0103	17.7500	2.7122	15.5000	18.7500
64	standing	dataset6.csv	5.0000	21.3300	17.9485	18.5000	2.5138	18.0000	19.3300
65	standing	dataset7.csv	0.0000	21.0000	15.0317	15.5850	4.3134	11.7500	18.5000
66	standing	dataset8.csv	6.5000	21.0000	16.8479	18.0000	2.6175	16.3300	18.2500
67	standing	dataset9.csv	1.5000	21.2500	17.6714	18.0000	1.7458	17.4575	18.3300
68	standing	dataset10.csv	6.3300	24.0000	16.3205	16.7500	2.4668	15.0000	18.0000
69	standing	dataset11.csv	1.0000	21.7500	13.2187	13.5000	4.7353	9.7300	17.5000
70	standing	dataset12.csv	0.0000	22.0000	11.7804	12.0000	3.0217	11.0000	13.3300
71	standing	dataset13.csv	0.0000	21.0000	12.1107	12.6700	3.5567	11.6275	14.3300
72	standing	dataset14.csv	0.0000	22.7500	12.7281	12.7100	3.8599	10.6700	15.0625
73	standing	dataset15.csv	3.0000	21.0000	13.3563	13.5000	2.9550	12.0000	15.5000
74	walking	dataset1.csv	5.5000	23.5000	15.7131	15.7500	2.8397	14.0000	17.6700
75	walking	dataset2.csv	5.7500	23.7500	15.5578	15.6350	2.9124	13.6275	17.5000
76	walking	dataset3.csv	5.0000	25.2500	15.2232	15.2500	3.1010	13.2500	17.3300
77	walking	dataset4.csv	5.6700	24.5000	15.4622	15.6700	3.1142	13.3300	17.5000
78	walking	dataset5.csv	5.0000	23.7500	15.2390	15.3300	2.9208	13.5000	17.2500
79	walking	dataset6.csv	6.5000	23.3300	15.5245	15.5000	2.9002	13.3300	17.6700
80	walking	dataset7.csv	7.6700	23.3300	15.5629	15.5000	2.9677	13.6700	17.5000
81	walking	dataset8.csv	5.5000	24.0000	15.1737	15.0000	3.0367	13.0000	17.3725
82	walking	dataset9.csv	7.6700	23.0000	15.6065	15.5000	2.8123	13.7500	17.5000
83	walking	dataset10.csv	6.5000	24.0000	15.2463	15.2500	2.9609	13.2500	17.2500
84	walking	dataset11.csv	7.3300	23.2500	15.3179	15.3300	2.9704	13.2500	17.2700
85	walking	dataset12.csv	6.2500	22.2500	15.2098	15.2500	2.7110	13.5000	17.0000
86	walking	dataset13.csv	7.0000	22.6700	15.1857	15.2500	2.7576	13.2500	17.0000
87	walking	dataset14.csv	6.6700	24.2500	15.4712	15.5000	2.8131	13.5000	17.2500
88	walking	dataset15.csv	6.0000	22.7500	15.4247	15.5000	2.8021	13.6700	17.3300

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 4, instances 1-44

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min4	max4	mean4	median4	std4	q1_4	q3_4
1	bending1	dataset1.csv	0.0000	7.2300	0.8325	0.5000	0.9657	0.0000	1.1200
2	bending1	dataset2.csv	0.0000	5.7600	0.8764	0.5000	1.0473	0.0000	1.1450
3	bending1	dataset3.csv	0.0000	4.4400	0.4973	0.4300	0.5507	0.0000	0.8300
4	bending1	dataset4.csv	0.0000	5.1500	0.9899	0.8300	0.9537	0.4300	1.3000
5	bending1	dataset5.csv	0.0000	6.4200	0.8419	0.5000	0.9288	0.4300	1.1200
6	bending1	dataset6.csv	0.0000	10.0300	0.8494	0.5000	0.9958	0.4300	1.1200
7	bending1	dataset7.csv	0.0000	5.1700	0.6664	0.4700	0.7890	0.0000	0.9400
8	bending2	dataset1.csv	0.0000	6.7600	0.7368	0.4700	0.9270	0.0000	0.8700
9	bending2	dataset2.csv	0.0000	5.7600	0.7799	0.5000	0.9044	0.0000	0.9400
10	bending2	dataset3.csv	0.0000	7.4000	0.8689	0.5000	1.0270	0.4300	1.1200
11	bending2	dataset4.csv	0.0000	4.4400	0.5446	0.4700	0.6444	0.0000	0.7100
12	bending2	dataset5.csv	0.0000	11.4200	0.7775	0.4700	1.0141	0.0000	1.1200
13	bending2	dataset6.csv	0.0000	5.3200	0.8235	0.5000	0.9022	0.0000	1.1200
14	cycling	dataset1.csv	0.0000	8.0500	2.9106	2.6200	1.6001	1.6400	3.9175
15	cycling	dataset2.csv	0.0000	8.3200	3.0339	2.9500	1.6254	1.7900	4.0300
16	cycling	dataset3.csv	0.0000	9.7400	2.9807	2.8600	1.6121	1.7900	4.0000
17	cycling	dataset4.csv	0.0000	9.5000	3.1191	2.9400	1.6835	1.7900	4.2175
18	cycling	dataset5.csv	0.0000	9.6300	2.9766	2.6850	1.8617	1.4800	4.0900
19	cycling	dataset6.csv	0.0000	10.5700	3.1385	2.8700	1.8477	1.7900	4.1100
20	cycling	dataset7.csv	0.0000	9.0100	3.0136	2.8600	1.7745	1.5600	4.0675
21	cycling	dataset8.csv	0.0000	8.0600	2.7841	2.5650	1.5430	1.5800	3.7700
22	cycling	dataset9.csv	0.0000	9.0000	2.8496	2.5000	1.7152	1.5800	3.9000
23	cycling	dataset10.csv	0.0000	10.6100	2.9370	2.6200	1.6615	1.7000	3.8700
24	cycling	dataset11.csv	0.0000	9.6300	3.0555	2.8700	1.7424	1.7975	4.0300
25	cycling	dataset12.csv	0.0000	9.4600	2.8711	2.6500	1.7048	1.5000	3.7800
26	cycling	dataset13.csv	0.0000	8.8700	3.1284	2.8700	1.7795	1.7225	4.3225
27	cycling	dataset14.csv	0.0000	9.7800	3.1298	2.8950	1.8105	1.7750	4.2400
28	cycling	dataset15.csv	0.0000	8.0500	2.9106	2.6200	1.6001	1.6400	3.9175
29	lying	dataset1.csv	0.0000	5.0200	0.8710	0.8200	0.7774	0.4300	1.2625
30	lying	dataset2.csv	0.0000	5.9100	0.5841	0.4300	0.7499	0.0000	0.7100
31	lying	dataset3.csv	0.0000	2.8600	0.3973	0.4300	0.4471	0.0000	0.5000
32	lying	dataset4.csv	0.0000	5.7200	0.3321	0.0000	0.5997	0.0000	0.4700
33	lying	dataset5.csv	0.0000	2.4900	0.3474	0.4300	0.4091	0.0000	0.5000
34	lying	dataset6.csv	0.0000	4.0300	0.8239	0.5000	0.7602	0.4300	1.1200
35	lying	dataset7.csv	0.0000	3.2700	0.3990	0.4300	0.4387	0.0000	0.5000
36	lying	dataset8.csv	0.0000	3.5400	0.5029	0.4700	0.5047	0.0000	0.7100
37	lying	dataset9.csv	0.0000	5.7200	0.7857	0.5000	0.8116	0.4300	1.0000
38	lying	dataset10.csv	0.0000	5.0200	0.8700	0.8200	0.7774	0.4300	1.2625
39	lying	dataset11.csv	0.0000	2.8600	0.3973	0.4300	0.4471	0.0000	0.5000
40	lying	dataset12.csv	0.0000	2.4900	0.3474	0.4300	0.4091	0.0000	0.5000
41	lying	dataset13.csv	0.0000	3.2700	0.3990	0.4300	0.4387	0.0000	0.5000
42	lying	dataset14.csv	0.0000	5.7200	0.7857	0.5000	0.8116	0.4300	1.0000
43	lying	dataset15.csv	0.0000	5.0200	0.8710	0.8200	0.7774	0.4300	1.2625
44	sitting	dataset1.csv	0.0000	5.3600	0.6419	0.4500	0.7610	0.0000	0.8700

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 4, instances 45-88

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min4	max4	mean4	median4	std4	q1_4	q3_4
45	sitting	dataset2.csv	0.0000	8.5000	0.6964	0.4700	0.8572	0.0000	0.8700
46	sitting	dataset3.csv	0.0000	6.3600	1.1087	0.8300	1.0153	0.4300	1.4800
47	sitting	dataset4.csv	0.0000	5.1700	0.6253	0.4700	0.6757	0.4300	0.8300
48	sitting	dataset5.csv	0.0000	5.1200	0.7405	0.4950	0.8329	0.0000	0.9400
49	sitting	dataset6.csv	0.0000	5.1200	0.6648	0.4700	0.7681	0.0000	0.8700
50	sitting	dataset7.csv	0.0000	6.0200	0.6543	0.4700	0.7083	0.0000	0.8700
51	sitting	dataset8.csv	0.0000	5.4500	0.5312	0.4300	0.7639	0.0000	0.7100
52	sitting	dataset9.csv	0.0000	4.6400	0.5102	0.4700	0.5427	0.0000	0.7200
53	sitting	dataset10.csv	0.0000	5.1200	0.8722	0.4950	0.9889	0.0000	1.2200
54	sitting	dataset11.csv	0.0000	6.9800	1.0477	0.8200	1.0970	0.4300	1.4800
55	sitting	dataset12.csv	0.0000	7.3100	0.8894	0.5000	1.0722	0.4300	0.9550
56	sitting	dataset13.csv	0.0000	5.6800	0.8996	0.8200	0.8362	0.4300	1.2500
57	sitting	dataset14.csv	0.0000	5.9300	0.6201	0.4700	0.7479	0.0000	0.8300
58	sitting	dataset15.csv	0.0000	6.1200	0.7583	0.4700	0.9077	0.0000	1.0000
59	standing	dataset1.csv	0.0000	5.7900	0.8431	0.5000	0.8888	0.4300	1.2200
60	standing	dataset2.csv	0.0000	6.5600	0.6867	0.4850	0.7653	0.0000	0.8700
61	standing	dataset3.csv	0.0000	5.8500	0.5889	0.4300	0.7010	0.0000	0.7100
62	standing	dataset4.csv	0.0000	7.5000	0.4680	0.4300	0.6872	0.0000	0.5000
63	standing	dataset5.csv	0.0000	6.0200	0.5443	0.4300	0.7437	0.0000	0.7100
64	standing	dataset6.csv	0.0000	5.5400	0.5662	0.4300	0.7010	0.0000	0.7100
65	standing	dataset7.csv	0.0000	5.1200	0.5985	0.4700	0.6792	0.0000	0.8300
66	standing	dataset8.csv	0.0000	6.5200	0.5354	0.4300	0.7133	0.0000	0.7100
67	standing	dataset9.csv	0.0000	4.1100	0.4413	0.4300	0.4782	0.0000	0.5000
68	standing	dataset10.csv	0.0000	5.5900	0.7499	0.5000	0.6676	0.4300	1.0375
69	standing	dataset11.csv	0.0000	5.6100	0.7999	0.5000	0.8678	0.4300	1.0200
70	standing	dataset12.csv	0.0000	4.7200	0.5642	0.4700	0.6382	0.0000	0.8300
71	standing	dataset13.csv	0.0000	5.5600	0.5675	0.4300	0.7184	0.0000	0.8200
72	standing	dataset14.csv	0.0000	3.7400	0.6427	0.4700	0.6773	0.0000	0.8700
73	standing	dataset15.csv	0.0000	5.5400	0.6456	0.4700	0.7182	0.0000	0.8300
74	walking	dataset1.csv	0.0000	8.8600	3.3038	3.2000	1.6673	2.0500	4.4400
75	walking	dataset2.csv	0.0000	9.1000	3.3509	3.1100	1.6644	2.1675	4.4400
76	walking	dataset3.csv	0.0000	8.5800	3.1113	2.8700	1.6887	1.8000	4.0900
77	walking	dataset4.csv	0.0000	8.7600	3.0729	2.8600	1.5087	1.9125	4.0900
78	walking	dataset5.csv	0.0000	9.2000	3.2071	3.0000	1.5905	2.0500	4.1950
79	walking	dataset6.csv	0.0000	9.0900	3.0928	2.8700	1.5034	2.0575	3.8400
80	walking	dataset7.csv	0.0000	9.0100	3.1983	2.9300	1.6763	2.0500	4.2450
81	walking	dataset8.csv	0.0000	9.1800	3.1455	3.0150	1.5913	2.0375	4.1575
82	walking	dataset9.csv	0.0000	9.2000	3.3654	3.0300	1.7308	2.1200	4.4400
83	walking	dataset10.csv	0.0000	9.5000	3.2791	3.1000	1.7594	2.0375	4.4025
84	walking	dataset11.csv	0.0000	9.0000	3.2272	3.1000	1.6475	2.0500	4.2600
85	walking	dataset12.csv	0.0000	9.0000	3.2772	3.1100	1.5000	2.1700	4.3900
86	walking	dataset13.csv	0.0000	8.3400	3.0287	2.8600	1.6652	1.7900	4.1100
87	walking	dataset14.csv	0.0000	9.9000	3.2057	3.0300	1.5193	2.0575	4.1900
88	walking	dataset15.csv	0.0000	9.1000	3.1883	3.0300	1.6025	2.0500	4.0375

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 5, instances 1-44

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min5	max5	mean5	median5	std5	q1_5	q3_5
1	bending1	dataset1.csv	27.2500	38.2500	34.3113	35.0000	2.1884	33.0000	36.0000
2	bending1	dataset2.csv	27.6700	38.5000	33.0246	33.0000	1.9953	32.0000	34.5000
3	bending1	dataset3.csv	29.0000	38.5000	35.5885	36.0000	1.9996	35.3625	36.5000
4	bending1	dataset4.csv	20.0000	38.6700	33.4939	35.0000	3.8494	30.4575	36.3300
5	bending1	dataset5.csv	23.6700	37.5000	29.8571	30.0000	2.4110	28.4575	31.2500
6	bending1	dataset6.csv	8.0000	33.5000	23.0348	23.5000	2.4889	22.2500	24.0000
7	bending1	dataset7.csv	11.3300	30.7500	22.1038	21.6700	3.3183	20.5000	23.7500
8	bending2	dataset1.csv	10.6700	30.0000	23.4930	23.7500	3.6938	20.5000	27.0000
9	bending2	dataset2.csv	0.0000	33.0000	17.6179	18.0000	5.0536	15.0000	20.7500
10	bending2	dataset3.csv	5.0000	33.7500	20.3457	19.5850	5.0324	17.6700	23.5000
11	bending2	dataset4.csv	11.0000	23.2500	18.1179	18.0000	1.7460	17.0000	19.0000
12	bending2	dataset5.csv	1.6700	36.0000	18.3891	17.5000	5.8459	15.0000	20.8125
13	bending2	dataset6.csv	0.0000	40.3300	14.4072	13.0000	7.8534	9.0000	18.3125
14	cycling	dataset1.csv	7.0000	25.5000	19.6073	20.0000	2.8903	17.9500	21.7500
15	cycling	dataset2.csv	10.5000	24.3300	19.5189	20.0000	2.7274	18.0000	21.5000
16	cycling	dataset3.csv	7.5000	24.2500	18.1260	18.8750	3.5371	16.0000	21.0000
17	cycling	dataset4.csv	2.5000	24.5000	15.8724	16.0000	3.1667	14.0000	18.0625
18	cycling	dataset5.csv	3.0000	24.6700	16.4800	16.7500	2.9782	14.6700	18.5000
19	cycling	dataset6.csv	6.7500	23.3300	16.4875	16.6700	2.8479	14.7500	18.5000
20	cycling	dataset7.csv	8.0000	23.0000	16.5811	16.7500	2.6559	15.0000	18.6700
21	cycling	dataset8.csv	2.0000	22.2500	15.9892	16.3300	2.8517	14.0000	18.2500
22	cycling	dataset9.csv	0.0000	23.0000	16.7735	17.0000	2.6893	15.0000	18.7500
23	cycling	dataset10.csv	8.6700	24.6700	17.2963	17.4150	2.7810	15.5000	19.2700
24	cycling	dataset11.csv	7.6700	24.5000	17.0626	16.7500	3.0881	15.0000	19.5000
25	cycling	dataset12.csv	4.0000	24.3300	16.0013	16.2500	3.1201	14.0000	18.0625
26	cycling	dataset13.csv	0.0000	26.5000	17.0783	17.0000	3.5376	14.7500	19.6900
27	cycling	dataset14.csv	7.6700	27.0000	18.4971	18.5000	3.6177	15.7500	21.0000
28	cycling	dataset15.csv	7.0000	25.5000	19.6073	20.0000	2.8903	17.9500	21.7500
29	lying	dataset1.csv	0.0000	21.0000	8.3374	8.7500	4.0745	5.5000	10.7500
30	lying	dataset2.csv	0.0000	12.7500	4.3763	3.3300	3.2745	2.0000	5.5425
31	lying	dataset3.csv	0.0000	12.0000	7.6249	9.0000	3.2685	4.6700	10.0000
32	lying	dataset4.csv	0.0000	22.2500	4.3040	2.2500	4.2538	1.0000	8.0000
33	lying	dataset5.csv	0.0000	14.5000	5.7795	5.7500	4.0974	1.6275	9.3300
34	lying	dataset6.csv	0.0000	16.2500	11.7656	12.0000	2.3367	11.3300	13.0000
35	lying	dataset7.csv	0.0000	18.0000	10.8975	11.5000	3.3216	9.0000	12.3300
36	lying	dataset8.csv	7.0000	18.6700	16.2123	16.7500	2.2234	15.7500	17.6700
37	lying	dataset9.csv	0.0000	21.0000	14.2991	14.0000	4.1421	11.7500	18.0000
38	lying	dataset10.csv	0.0000	21.0000	8.3370	8.7500	4.0745	5.5000	10.7500
39	lying	dataset11.csv	0.0000	12.0000	7.6249	9.0000	3.2685	4.6700	10.0000
40	lying	dataset12.csv	0.0000	14.5000	5.7795	5.7500	4.0974	1.6275	9.3300
41	lying	dataset13.csv	0.0000	18.0000	10.8975	11.5000	3.3216	9.0000	12.3300
42	lying	dataset14.csv	0.0000	21.0000	14.2991	14.0000	4.1421	11.7500	18.0000
43	lying	dataset15.csv	0.0000	21.0000	8.3374	8.7500	4.0745	5.5000	10.7500
44	sitting	dataset1.csv	1.0000	28.0000	13.3818	13.1250	4.9568	10.1875	17.6900

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 5, instances 45-88

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min5	max5	mean5	median5	std5	q1_5	q3_5
45	sitting	dataset2.csv	2.0000	24.0000	18.4428	19.3300	3.5855	16.5000	21.0000
46	sitting	dataset3.csv	4.7500	26.7500	19.3307	20.0000	2.9458	17.6275	21.0000
47	sitting	dataset4.csv	4.3300	30.7500	19.2651	19.0000	4.1275	17.5000	21.6900
48	sitting	dataset5.csv	5.5000	27.0000	17.1497	18.0000	2.8636	15.5000	19.0000
49	sitting	dataset6.csv	2.0000	23.5000	18.0872	19.2500	3.7363	16.5000	20.7500
50	sitting	dataset7.csv	0.0000	23.7500	16.5220	17.7500	4.9785	14.0000	20.7500
51	sitting	dataset8.csv	1.0000	23.0000	16.0257	17.0000	4.1123	13.2500	19.0000
52	sitting	dataset9.csv	5.6700	22.0000	16.0432	16.0000	3.0272	14.0000	18.3300
53	sitting	dataset10.csv	0.0000	27.3300	15.1673	15.2500	5.3138	11.6700	19.7500
54	sitting	dataset11.csv	1.5000	25.0000	14.9205	15.8750	5.2586	10.5000	19.2500
55	sitting	dataset12.csv	1.0000	23.0000	12.9739	13.2900	4.8475	9.0000	17.2500
56	sitting	dataset13.csv	0.0000	23.2500	12.5372	12.0000	6.0398	8.5000	18.2500
57	sitting	dataset14.csv	2.5000	26.6700	15.2604	13.2500	6.4668	9.7500	22.0000
58	sitting	dataset15.csv	2.0000	24.0000	16.4089	17.6700	4.2940	13.7300	19.0000
59	standing	dataset1.csv	0.0000	23.0000	13.4437	14.7500	5.4018	9.3300	17.7500
60	standing	dataset2.csv	2.0000	21.2500	14.2855	14.6700	2.9840	12.7500	16.5000
61	standing	dataset3.csv	1.3300	21.3300	15.5508	16.5850	4.2966	13.0000	18.5650
62	standing	dataset4.csv	1.0000	20.7500	12.0604	12.2900	3.1417	10.6275	14.2500
63	standing	dataset5.csv	1.0000	20.2500	13.1983	13.7500	3.2891	11.3100	15.5425
64	standing	dataset6.csv	1.0000	19.6700	13.2101	13.0000	2.6124	12.0000	14.8125
65	standing	dataset7.csv	2.5000	21.0000	13.3940	13.5000	2.9316	12.0000	15.2500
66	standing	dataset8.csv	2.6700	21.2500	13.2819	13.6700	3.1348	11.6700	15.5000
67	standing	dataset9.csv	1.0000	19.7500	12.4285	13.1250	3.3741	11.2500	14.5000
68	standing	dataset10.csv	2.0000	22.0000	10.0677	9.7500	3.7221	7.6275	12.0000
69	standing	dataset11.csv	1.0000	21.0000	14.6397	15.0000	3.6236	12.6275	17.5000
70	standing	dataset12.csv	0.0000	20.2500	14.9460	15.2500	2.7026	14.0000	16.6900
71	standing	dataset13.csv	0.0000	19.6700	14.2475	14.5000	3.2616	12.7500	16.5000
72	standing	dataset14.csv	2.0000	24.0000	18.2014	18.2500	3.5660	16.5000	21.0000
73	standing	dataset15.csv	1.0000	21.0000	12.6115	12.7500	3.4145	11.0000	14.6700
74	walking	dataset1.csv	6.7500	26.0000	16.6212	16.6700	3.0921	14.7500	18.6700
75	walking	dataset2.csv	7.2500	25.0000	16.5414	16.7500	3.1336	14.6275	18.7500
76	walking	dataset3.csv	7.0000	24.5000	16.2510	16.3300	3.1550	14.2500	18.5000
77	walking	dataset4.csv	6.2500	23.5000	16.1012	16.3300	3.2076	14.2500	18.5000
78	walking	dataset5.csv	6.0000	25.5000	16.2996	16.2500	3.1747	14.2500	18.3300
79	walking	dataset6.csv	6.3300	25.0000	15.9972	16.2500	3.1921	14.2375	18.2500
80	walking	dataset7.csv	6.6700	24.0000	15.8644	16.0000	3.0005	13.7500	18.0000
81	walking	dataset8.csv	6.5000	23.2500	16.0556	16.0000	2.9057	14.0000	18.2500
82	walking	dataset9.csv	5.7500	24.0000	16.1480	16.2500	2.9929	14.3300	18.2500
83	walking	dataset10.csv	7.0000	24.2500	15.7161	15.7500	3.1166	13.7500	18.0000
84	walking	dataset11.csv	6.3300	25.0000	15.8921	16.0000	3.2840	13.7300	18.2500
85	walking	dataset12.csv	6.2500	23.2500	15.5505	15.7500	3.1199	13.5000	17.7500
86	walking	dataset13.csv	7.0000	22.7500	15.7595	15.7500	2.8231	14.0000	17.7500
87	walking	dataset14.csv	5.5000	23.5000	15.8739	16.0000	3.1311	13.7500	18.0000
88	walking	dataset15.csv	4.6700	23.2500	15.7042	16.0000	3.1563	13.7300	17.7500

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 6, instances 1-44

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min6	max6	mean6	median6	std6	q1_6	q3_6
1	bending1	dataset1.csv	0.0000	1.9200	0.5706	0.4300	0.5829	0.0000	1.3000
2	bending1	dataset2.csv	0.0000	3.1100	0.5711	0.4300	0.6010	0.0000	1.3000
3	bending1	dataset3.csv	0.0000	1.7900	0.4933	0.4300	0.5135	0.0000	0.9400
4	bending1	dataset4.csv	0.0000	2.1800	0.6135	0.5000	0.5243	0.0000	1.0000
5	bending1	dataset5.csv	0.0000	1.7900	0.3833	0.4300	0.3892	0.0000	0.5000
6	bending1	dataset6.csv	0.0000	5.2600	0.6796	0.5000	0.6225	0.4300	0.8700
7	bending1	dataset7.csv	0.0000	2.9600	0.5553	0.4900	0.4878	0.0000	0.8300
8	bending2	dataset1.csv	0.0000	4.9700	0.7002	0.5000	0.6937	0.4300	0.8700
9	bending2	dataset2.csv	0.0000	6.7600	1.1221	0.8300	1.0123	0.4700	1.3000
10	bending2	dataset3.csv	0.0000	13.6100	1.1620	0.8300	1.3330	0.4700	1.3000
11	bending2	dataset4.csv	0.0000	6.4000	0.7016	0.7100	0.4811	0.4700	0.9400
12	bending2	dataset5.csv	0.0000	6.7300	1.1074	0.8300	1.0808	0.4700	1.3000
13	bending2	dataset6.csv	0.0000	4.9200	1.0981	0.9400	0.8315	0.5000	1.3000
14	cycling	dataset1.csv	0.0000	9.3400	2.9217	2.5000	1.8526	1.5000	3.9000
15	cycling	dataset2.csv	0.0000	9.6200	2.7659	2.4500	1.7692	1.4100	3.7700
16	cycling	dataset3.csv	0.0000	8.5500	2.9838	2.5700	1.8157	1.5000	4.1500
17	cycling	dataset4.csv	0.0000	9.9800	3.4807	3.3400	1.8278	2.1025	4.5500
18	cycling	dataset5.csv	0.0000	8.1900	3.0733	2.6900	1.6297	1.9125	4.0875
19	cycling	dataset6.csv	0.0000	9.5000	3.0764	2.7700	1.8245	1.7000	4.0375
20	cycling	dataset7.csv	0.0000	8.8100	2.7733	2.5900	1.5699	1.6400	3.6325
21	cycling	dataset8.csv	0.0000	8.3400	2.9346	2.5250	1.6314	1.6600	4.0300
22	cycling	dataset9.csv	0.0000	8.7500	2.8224	2.5900	1.6372	1.5800	3.7400
23	cycling	dataset10.csv	0.0000	8.9900	2.8876	2.5250	1.7231	1.5600	3.7700
24	cycling	dataset11.csv	0.0000	9.1800	3.2255	2.8700	1.7698	1.8850	4.2625
25	cycling	dataset12.csv	0.0000	9.3900	3.0697	2.7700	1.7483	1.7975	4.0600
26	cycling	dataset13.csv	0.0000	8.5000	3.0930	2.9300	1.6260	1.8900	4.0600
27	cycling	dataset14.csv	0.0000	11.1500	3.5305	3.1100	1.9637	2.1700	4.6175
28	cycling	dataset15.csv	0.0000	9.3400	2.9217	2.5000	1.8526	1.5000	3.9000
29	lying	dataset1.csv	0.0000	4.5000	0.7343	0.7100	0.6137	0.4300	1.0000
30	lying	dataset2.csv	0.0000	3.9100	0.6928	0.5000	0.6758	0.3225	0.9400
31	lying	dataset3.csv	0.0000	2.5000	0.6412	0.5000	0.3884	0.4600	0.8300
32	lying	dataset4.csv	0.0000	4.9700	0.5493	0.4700	0.6484	0.0000	0.8300
33	lying	dataset5.csv	0.0000	3.4900	0.6359	0.5000	0.6084	0.0000	0.8300
34	lying	dataset6.csv	0.0000	3.1900	0.6229	0.5000	0.4989	0.4300	0.8700
35	lying	dataset7.csv	0.0000	4.0600	0.5827	0.5000	0.4384	0.4300	0.8300
36	lying	dataset8.csv	0.0000	3.5000	0.5885	0.5000	0.4234	0.4300	0.8300
37	lying	dataset9.csv	0.0000	5.7200	0.7662	0.5000	0.7240	0.4300	1.0000
38	lying	dataset10.csv	0.0000	4.5000	0.7354	0.7100	0.6128	0.4300	1.0000
39	lying	dataset11.csv	0.0000	2.5000	0.6412	0.5000	0.3884	0.4600	0.8300
40	lying	dataset12.csv	0.0000	3.4900	0.6359	0.5000	0.6084	0.0000	0.8300
41	lying	dataset13.csv	0.0000	4.0600	0.5827	0.5000	0.4384	0.4300	0.8300
42	lying	dataset14.csv	0.0000	5.7200	0.7662	0.5000	0.7240	0.4300	1.0000
43	lying	dataset15.csv	0.0000	4.5000	0.7343	0.7100	0.6137	0.4300	1.0000
44	sitting	dataset1.csv	0.0000	6.0200	1.0340	0.8300	0.9856	0.4700	1.2500

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.

1(c)(ii) Appendix: time series 6, instances 45-88

All extracted values, rounded to four decimals. Instance IDs identify rows consistently across channels; the supplied file and activity identify each recording. Full precision and split labels are in results/extracted_features.csv.

ID	Activity	File	min6	max6	mean6	median6	std6	q1_6	q3_6
45	sitting	dataset2.csv	0.0000	7.2200	0.9046	0.7100	0.8962	0.4300	1.1200
46	sitting	dataset3.csv	0.0000	8.2000	0.8372	0.7100	0.8353	0.4300	1.1200
47	sitting	dataset4.csv	0.0000	4.3000	0.7641	0.7000	0.6521	0.4300	1.0000
48	sitting	dataset5.csv	0.0000	6.2600	0.8548	0.8300	0.7176	0.4300	1.1450
49	sitting	dataset6.csv	0.0000	5.1200	0.7914	0.7100	0.7557	0.4300	1.0000
50	sitting	dataset7.csv	0.0000	5.1700	0.8402	0.8200	0.7506	0.4300	1.0900
51	sitting	dataset8.csv	0.0000	6.3600	0.8560	0.8200	0.7648	0.4300	1.0900
52	sitting	dataset9.csv	0.0000	5.4500	0.8062	0.8200	0.6455	0.4300	1.0900
53	sitting	dataset10.csv	0.0000	6.3600	1.0577	0.8300	1.0259	0.4700	1.3000
54	sitting	dataset11.csv	0.0000	7.0000	1.3549	1.0900	1.1509	0.5000	1.7000
55	sitting	dataset12.csv	0.0000	6.3600	0.9612	0.8200	0.8828	0.4700	1.2200
56	sitting	dataset13.csv	0.0000	7.8500	0.8690	0.8200	0.7942	0.4700	1.0000
57	sitting	dataset14.csv	0.0000	4.6400	0.7198	0.5000	0.6085	0.4300	0.8300
58	sitting	dataset15.csv	0.0000	5.1000	0.8921	0.7100	0.8512	0.4300	1.2200
59	standing	dataset1.csv	0.0000	5.0200	0.9330	0.8300	0.6736	0.4700	1.2500
60	standing	dataset2.csv	0.0000	5.7200	0.9120	0.8300	0.6662	0.4700	1.2200
61	standing	dataset3.csv	0.0000	5.7300	0.8423	0.7100	0.7222	0.4300	1.0900
62	standing	dataset4.csv	0.0000	4.6400	0.9174	0.8300	0.7096	0.4700	1.1200
63	standing	dataset5.csv	0.0000	6.1800	1.0397	0.8300	0.9167	0.4700	1.2200
64	standing	dataset6.csv	0.0000	4.3200	0.9274	0.8300	0.7564	0.4700	1.2200
65	standing	dataset7.csv	0.0000	6.0000	0.8826	0.8300	0.6684	0.4700	1.1200
66	standing	dataset8.csv	0.0000	6.5800	0.9911	0.8300	0.8553	0.4700	1.2200
67	standing	dataset9.csv	0.0000	4.5000	0.7951	0.8200	0.5030	0.4700	1.0000
68	standing	dataset10.csv	0.0000	6.6500	1.2263	1.0900	0.8920	0.5000	1.5850
69	standing	dataset11.csv	0.0000	6.8500	0.9774	0.8300	0.8533	0.4700	1.2200
70	standing	dataset12.csv	0.0000	4.0000	0.7485	0.8200	0.4612	0.4300	0.9500
71	standing	dataset13.csv	0.0000	3.7700	0.7020	0.5000	0.5675	0.4300	0.9400
72	standing	dataset14.csv	0.0000	3.8300	0.6455	0.5000	0.5674	0.4300	0.8300
73	standing	dataset15.csv	0.0000	5.9100	1.1551	0.9400	0.8421	0.5000	1.5000
74	walking	dataset1.csv	0.0000	9.7400	3.3941	3.1000	1.7921	2.1050	4.4250
75	walking	dataset2.csv	0.0000	8.9600	3.3785	3.0850	1.7874	2.0600	4.4400
76	walking	dataset3.csv	0.0000	8.9900	3.2444	3.0000	1.6310	2.1200	4.2400
77	walking	dataset4.csv	0.0000	8.5000	3.2420	3.0150	1.7692	1.8850	4.4400
78	walking	dataset5.csv	0.0000	9.3900	3.2883	3.2700	1.6475	2.0500	4.3050
79	walking	dataset6.csv	0.0000	10.2100	3.2800	3.0150	1.7009	2.1200	4.5000
80	walking	dataset7.csv	0.0000	8.0100	3.2616	2.9800	1.6173	2.0500	4.3200
81	walking	dataset8.csv	0.0000	8.8600	3.2895	3.0150	1.6802	2.1200	4.2600
82	walking	dataset9.csv	0.0000	9.4200	3.4795	3.2700	1.7611	2.2400	4.5375
83	walking	dataset10.csv	0.0000	8.3200	3.5008	3.2850	1.6924	2.1800	4.5575
84	walking	dataset11.csv	0.0000	8.3200	3.2597	3.1100	1.6402	2.0500	4.3225
85	walking	dataset12.csv	0.0000	9.6700	3.4326	3.2000	1.7327	2.1575	4.5650
86	walking	dataset13.csv	0.0000	10.0000	3.3381	3.0800	1.6567	2.1600	4.3350
87	walking	dataset14.csv	0.0000	9.5100	3.4246	3.2700	1.6910	2.1700	4.5000
88	walking	dataset15.csv	0.4300	9.0000	3.3405	3.0900	1.6991	2.1200	4.3750

Definitions: std = sample standard deviation (ddof = 1); q1 and q3 = 25th and 75th percentiles with linear interpolation. sitting/dataset8.csv uses 479 observed points; every other file uses 480.